



The University of Manchester

An Introduction to Uncertainty Quantification:

Foundations and applications to predictive models

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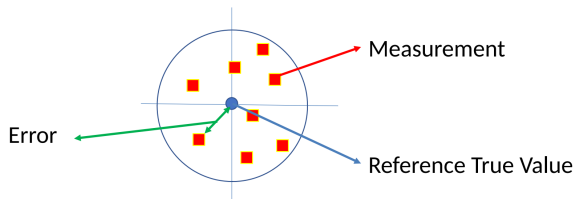
- Part I: Foundation of UQ
- Part II: Gaussian processes
- Part III: UQ for neural network models
- Part IV: Trustworthy UQ, risk assessment, and decision making

Part I

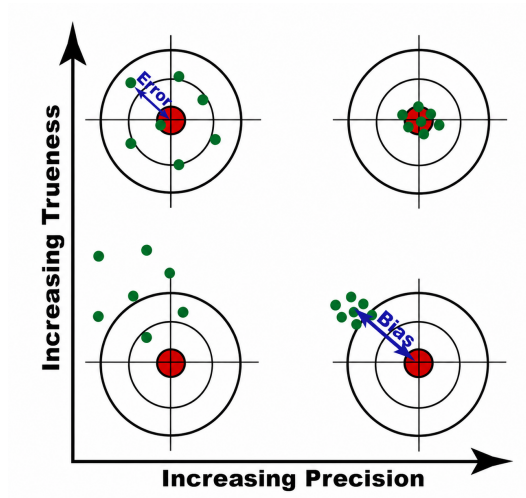
Foundation of Uncertainty Quantification

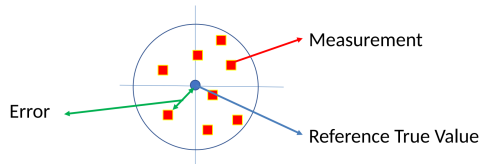
1. Basic Concepts
2. Verification, Validation & UQ (VVUQ)
3. Fundamental UQ Problems

- **True value:** “the value of a quantity that, were it known, would ideally reflect the property of an object with respect to the purpose of the measurement” [11].
 - In the real world, we can get **reference true value**, at best.
- **Error:** Deviation of a realization of a QoI from the ground truth.
- **Uncertainty:** An estimated range of values within which the true value of the QoI exists.



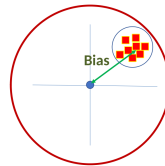
⚠ **Uncertainty $\neq$ Error**





Aleatoric (Random)

- Irreducible
- Inherent
- Naturally defined in a statistical framework



Epistemic (Biased)

- Reducible
- Lack of knowledge, missing physics, model imperfection, approximations
- Less/not naturally defined in a statistical framework

✔ Sometimes, this distinction is difficult to make. ✔ We may deliberately represent one type of uncertainty in terms of the other.

Examples often modelled as aleatory uncertainties:

- **Manufacturing Tolerances:** Variations in fuel rod diameter, cladding thickness, pellet enrichment, and enrichment distribution.
- **Geometric Displacements:** Rod bowing, spacer grid misalignment, or pitch variations caused by thermal expansion and irradiation.
- **Operational Fluctuations:** Minor shifts in total core thermal power, inlet temperature, system pressure, and total core flow rate.

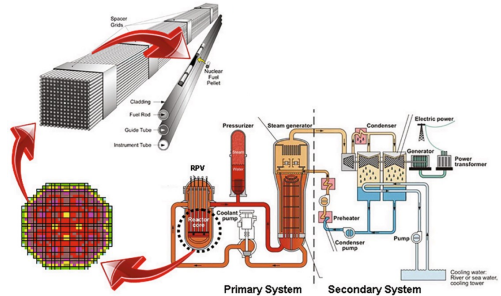
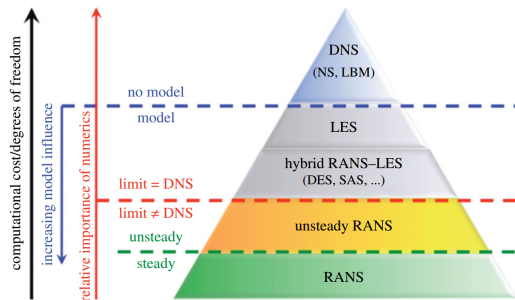


Figure from [Kothe 2010](#).

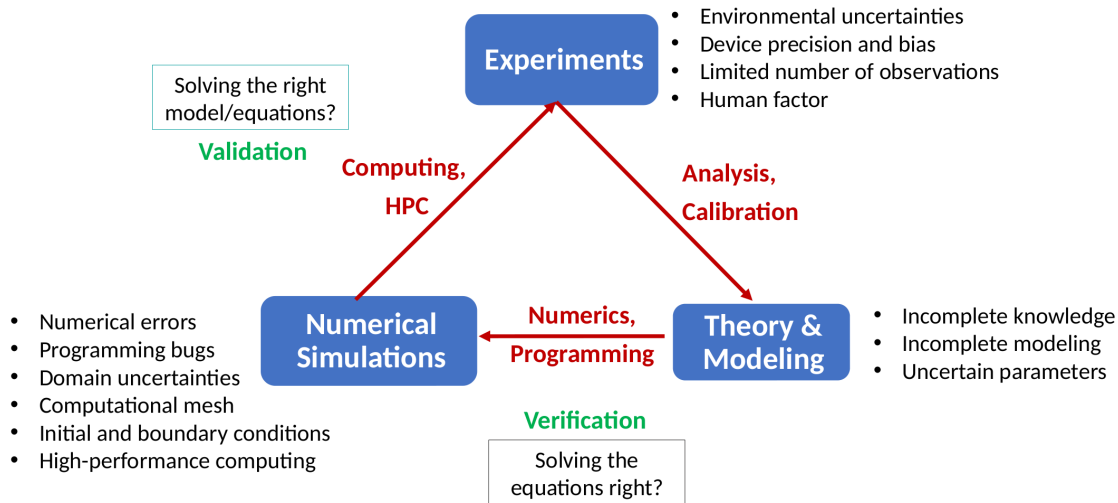
Examples often treated as epistemic uncertainties:

- **Empirical Correlations:** These include subchannel-to-subchannel turbulent mixing coefficients, two-phase friction multipliers, void drift parameters, and critical heat flux (CHF) lookup tables.
- **Numerical Approximations:** Uncertain behaviour or values of the errors stemming from spatial mesh discretization and time-step sizing.
- **Turbulence modelling and simulation:** Presence of modelling, numerical, sampling, initial/boundary conditions, etc uncertainties.



Hierarchy of turbulence modelling & simulation approaches

1. Basic Concepts
2. Verification, Validation & UQ (VVUQ)
3. Fundamental UQ Problems



1. Basic Concepts
2. Verification, Validation & UQ (VVUQ)
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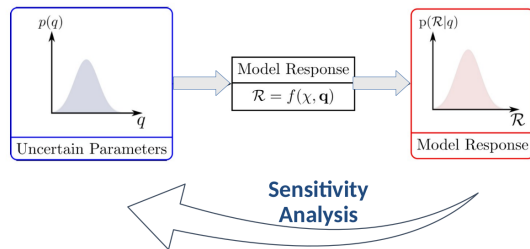
R. Smith, 2024 [16]: "UQ can be broadly defined as the science of identifying, quantifying, and reducing uncertainties associated with models, numerical algorithms, experiments, and predicted QoI (quantity of interest)."

Required steps:

1. Define/identify the UQ problem
2. Identify type/sources of uncertainties (in data, mathematical models, inputs, parameters, etc)
3. Specify QoIs
4. Adopt appropriate UQ techniques
5. Interpret the UQ results.

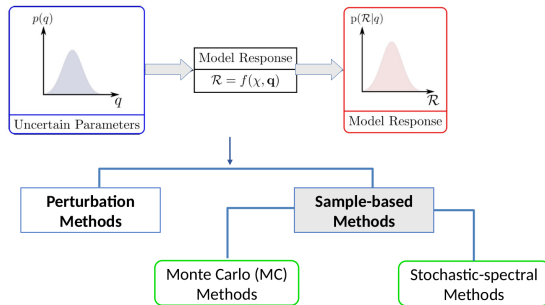
Model: $y = f(\mathbf{q})$ with $\mathbf{q}$: uncertain inputs

- **Uncertainty propagation:** What is the distribution of Qols, given the distribution of uncertain inputs?
- **Global Sensitivity Analysis:** What are the contributions of uncertain inputs to the propagated uncertainty in the Qols? (e.g. Sobol indices [17])
- Instead of distribution, sometimes, we are only interested in the **stochastic moments** of the Qols:



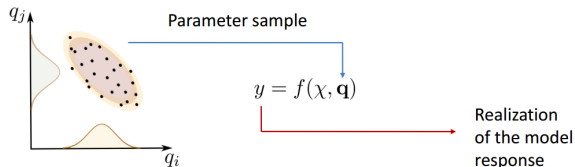
$$\begin{aligned}\mathbb{E}[y] &= \int_{\mathbf{Q}} f(\mathbf{q}) \rho(\mathbf{q}) d\mathbf{q} \\ \mathbb{V}[y] &= \int_{\mathbf{Q}} (f(\mathbf{q}) - \mathbb{E}[y])^2 \rho(\mathbf{q}) d\mathbf{q}\end{aligned}$$

- **Perturbation method:** used by engineers; only an approximation; use with caution for non-small uncertainty and non-Gaussian variables.
- **Sample-based methods:** can be used for obtaining the distribution and stochastic moments of the Qols.



Non-intrusive sample-based methods:

1. Draw samples from uncertain inputs.
2. For each sample, evaluate Qol by running the simulator.
3. Collect dataset $\mathcal{D} = \{(\mathbf{q}_i, y_i)\}_{i=1}^n$
4. If $f(\cdot)$ is not known, construct a surrogate: $y = \tilde{f}(\mathbf{q})$.
5. Find distribution or stochastic moments the output using the model or its surrogate.



Non-intrusive Monte Carlo method:

1. Draw samples $\{\mathbf{q}_i\}_{i=1}^n$ from the joint PDF $\rho(\mathbf{q})$
2. Evaluate Qols by running the simulator $y_i = f(\mathbf{q}_i)$
3. Estimate the stochastic moments directly from integrals.

Pros:

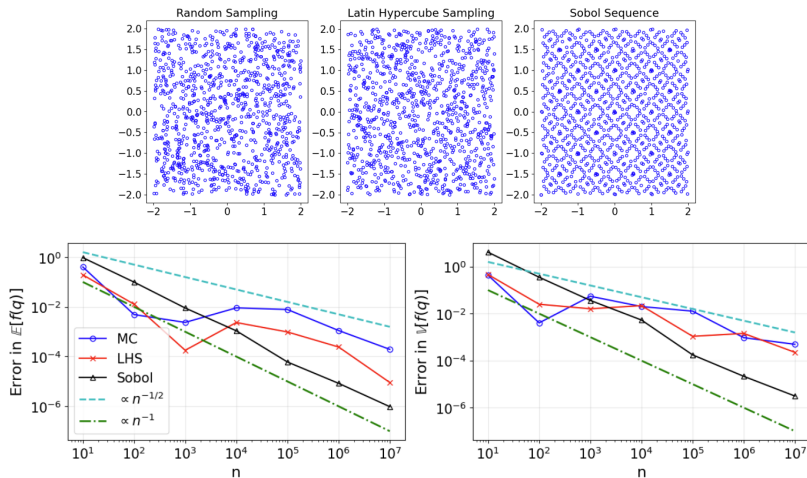
- Easy to implement – needs an automated workflow
- Integration convergence is independent of the dimensionality of $\mathbf{q}$

Cons:

- Low convergence rate of integrals:
pure MC: $\mathcal{O}(n^{-1/2})$; Quasi-MC: At most $\mathcal{O}(n^{-1})$

Estimate stochastic mean and variance of the Rosenbrock function:

$$f(\mathbf{q}) = f(q_1, q_2) = 100(q_2 - q_1^2)^2 + (1 - q_1)^2, \quad q_1, q_2 \sim \mathcal{U}[-2, 2]$$



Polynomial Chaos Expansion (PCE) is the spectral representation of a stochastic process:

$$\tilde{f}(\mathbf{q}) = \sum_{k=0}^K \hat{f}_k \Psi_k(\mathbf{q})$$

where $\Psi_k(\mathbf{q})$ and $\Psi_m(\mathbf{q})$ are orthogonal with respect to the joint PDF $\rho(\mathbf{q})$:

$$\langle \Psi_k(\mathbf{q}), \Psi_m(\mathbf{q}) \rangle_\rho = \int \Psi_k(\mathbf{q}) \Psi_m(\mathbf{q}) \rho(\mathbf{q}) d\mathbf{q} = \|\Psi_k\|_2^2 \delta_{mk}$$

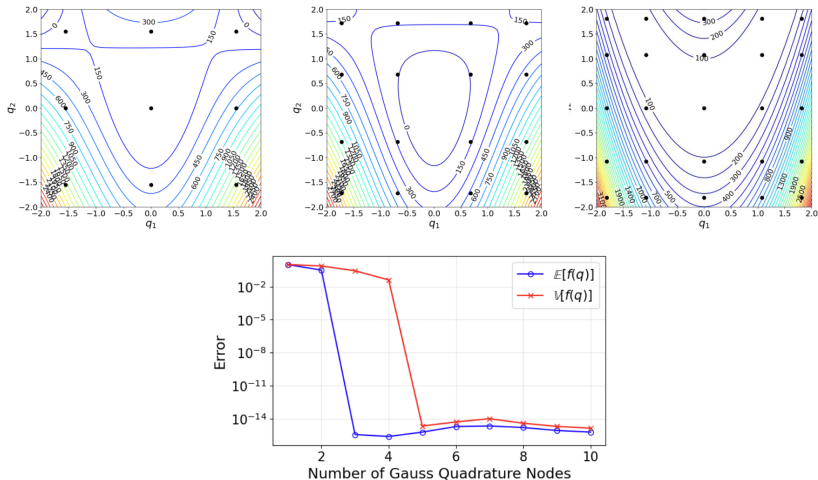
Non-intrusive PCE:

1. Based on the distribution of $\{q_i\}_{i=1}^P$, choose! [18] or construct $\psi_{k_i}(q_i)$.
2. Create a set of training data $\mathcal{D}$; samples of $\mathbf{q}$ may depend on the bases.
3. Given the bases, estimate coefficients $\{\hat{f}_k\}_{k=0}^K$ using Regression or Projection methods.
4. Analytical expressions for stochastic moments:

$$\begin{aligned} \mathbb{E}[f(\mathbf{q})] &= \hat{f}_0 \\ \mathbb{V}[f(\mathbf{q})] &= \sum_{k=1}^K \hat{f}_k^2 \|\Psi_k\|_2^2 \end{aligned}$$

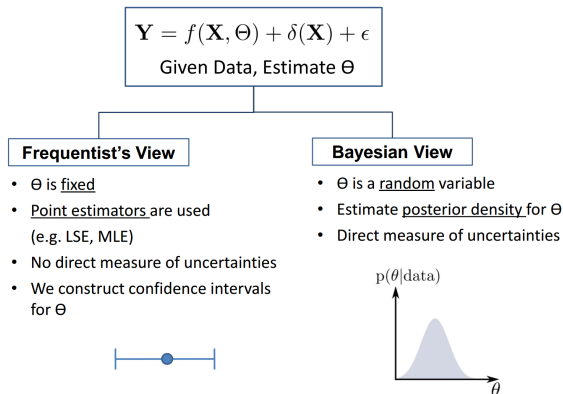
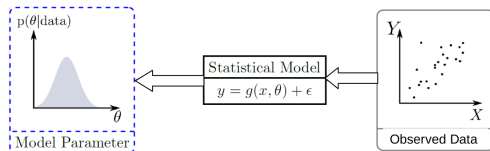
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Use (noisy) data $\mathcal{D}$ to

- Infer $g(x; \theta)$, or
- Infer θ , for a given structure of $g(x, \theta)$.



$$Y_i = f(X_i, \bar{\Theta}) + \varepsilon_i, \quad i = 1, 2, \dots, n \quad \text{where} \quad \mathbb{V}(\varepsilon) = \bar{\sigma}^2$$

$\bar{\Theta}, \bar{\sigma}^2$: Fixed but unknown $\implies$ Need point estimators to find “best” parameter estimate

Least Squares Estimator (LSE):

Find the parameters which minimize the L_2 -norm of the misfit between data and model

$$\hat{\Theta}_{\text{LSE}} = \min_{\Theta} \sum_{i=1}^n [Y_i - f(X_i, \Theta)]^2$$

$$\begin{cases} \mathcal{J}(\Theta) = (\mathbf{Y} - f(\mathbf{X}, \Theta))^T (\mathbf{Y} - f(\mathbf{X}, \Theta)) \\ \hat{\theta}_{\text{LSE}} = \min_{\Theta} \mathcal{J}(\Theta) \end{cases}$$

- Do additional calculations to construct a confidence interval for $\hat{\theta}_{\text{LSE}}$.
- The above step requires linearisation of non-linear $f(x, \theta)$ with respect to θ ($\Rightarrow$ approximate intervals).

⚠ Note: Maximum Likelihood Estimator (MLE) is a point estimator. Under certain assumptions it is the same as LSE (Proof in the Gaussian Process Regression lecture).

Bayes' Theorem:

$$p(\theta|\mathcal{D}) = \frac{p(\mathcal{D}|\theta) p_0(\theta)}{p(\mathcal{D})}$$

- $p_0(\theta)$: Prior distribution of θ (chosen)
- $p(\mathcal{D}|\theta)$: Likelihood (constructed)
- $p(\mathcal{D})$: Marginal likelihood or evidence

$$p(\mathcal{D}) = \int_{\Omega_\theta} p(\mathcal{D}|\theta) p_0(\theta) d\theta$$

- $p(\theta|\mathcal{D})$: Posterior distribution of θ given data $\mathcal{D}$
→ to be inferred!

Inference of $p(\theta|\mathcal{D})$:[†]

1. Markov chain Monte Carlo (MCMC) methods
2. Bayesian Variational Methods

[†] Suggested references: Bardsley 2018 [2]; Reich-Ghosh 2026 [13].

MCMC method:

- Construct a Markov chain s.t. its stationary distribution is the posterior density.
- Samples will be taken from the posterior density. Once we have enough number of samples, we can estimate the posterior density with some accuracy.
- There are various algorithms for MCMC, see e.g. [5, 8, 6]. These include Metropolis, Metropolis-Hasting, Adaptive and delayed rejection Metropolis-Hasting algorithms.

Metropolis-Hasting Algorithm: (See the history and details in [14])

Assume Θ_0

do $k = 1, n$:

- 1 sample $\Theta^* \sim R(\Theta^* | \Theta_{k-1})$  R: proposal (jumping) distribution
- 2 compute $r = p(\Theta^* | \mathcal{D}) R(\Theta_{k-1} | \Theta^*) / p(\Theta_{k-1} | \mathcal{D}) R(\Theta^* | \Theta_{k-1})$
- 3 accept or reject the proposal:

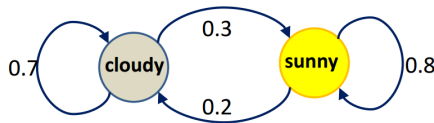
$$\Theta_k = \begin{cases} \Theta^*, & \text{with probability } \min(1, r) \\ \Theta_{k-1}, & \text{otherwise} \end{cases}$$

Markov chain: A sequence of states where the new state depends only on the current state (memory-less).

1. State Space = [cloudy , sunny]

2. Transition Matrix
(probability of changing state)

$$\mathbf{P} = \begin{bmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{bmatrix}$$



3. Initial Distribution (Today state): $\mathbf{p}_0 = [1, 0]$

$$\mathbf{p}_1 = \mathbf{p}_0 \mathbf{P} = [0.7, 0.3]$$

$$\mathbf{p}_2 = \mathbf{p}_1 \mathbf{P} = \mathbf{p}_0 \mathbf{P}^2 = [0.55, 0.45]$$

Predictions: $\mathbf{p}_3 = \mathbf{p}_2 \mathbf{P} = \mathbf{p}_0 \mathbf{P}^3 = [0.475, 0.525]$

$$\mathbf{p}_4 = \mathbf{p}_3 \mathbf{P} = \mathbf{p}_0 \mathbf{P}^4 = [0.4375, 0.5625]$$

$\vdots$

$$\mathbf{p}_{10} = \mathbf{p}_9 \mathbf{P} = \mathbf{p}_0 \mathbf{P}^{10} = [0.4006, 0.5994]$$

Stationary Distribution

$$\mathbf{p}_s \mathbf{P} = \mathbf{p}_s$$

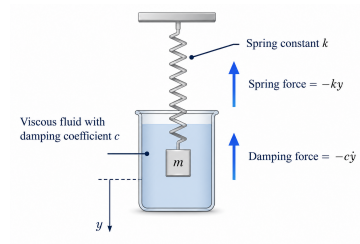
No more
change

Consider a damped harmonic oscillator:

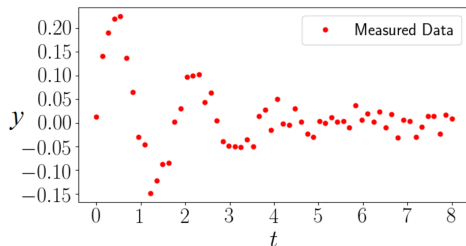
$$\begin{cases} \ddot{y} + c\dot{y} + ky = 0 \\ y(0) = y_0, \dot{y}(0) = v_0 \end{cases}$$

$$y(t) = \exp\left(-\frac{c}{2}t\right)(c_1 \cos \omega t + c_2 \sin \omega t)$$

$$\omega = \sqrt{k - c^2/4}, c_1 = y_0, c_2 = (v_0 + cy_0/2)/\omega$$

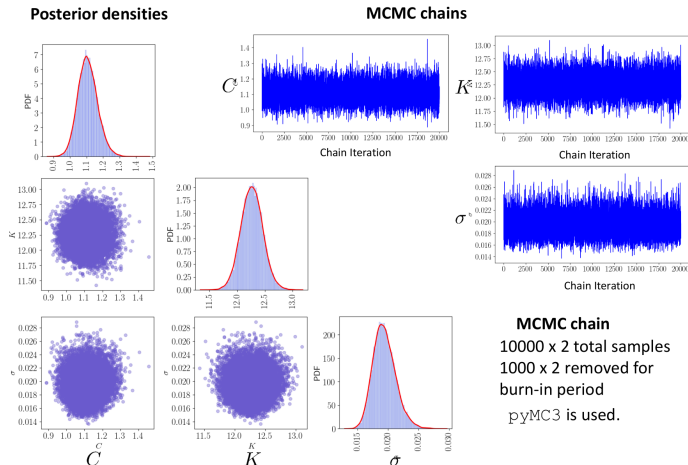


Given a set of noisy observations data $\mathcal{D}$, estimate k and c .



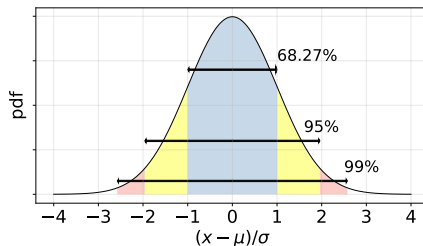
Assume: $y_{obs_i} = y(t_i, \theta) + \varepsilon_i$, $\varepsilon_i = \mathcal{N}(0, \sigma^2) \Rightarrow \theta = \{k, c\}$

Construct the likelihood: $p(\mathcal{D}|\theta, \sigma) = L(\theta, \sigma|\mathcal{D}) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_{obs_i} - y(t_i, \theta))^2\right)$



- Sometimes, we summarise uncertainty information as an interval. But be careful how to interpret it!
- Suppose our model predicts the QoI to be approximately Gaussian: $Y \sim \mathcal{N}(\mu, \sigma^2)$.
- Then, $[\mu - c\sigma, \mu + c\sigma]$ contains $\alpha\%$ of possible realizations of the QoI.

$\alpha\%$	c
68.3%	1.00
95%	1.96
99%	2.58



Interpretation: A 95% interval

- ✓ does NOT mean “95% certainty that this particular realization lies inside the range.”
- ✓ means across many realizations of the QoI, approximately 95% of them will fall inside such interval.

Part II

Gaussian processes for surrogate construction

See the GPR slides!

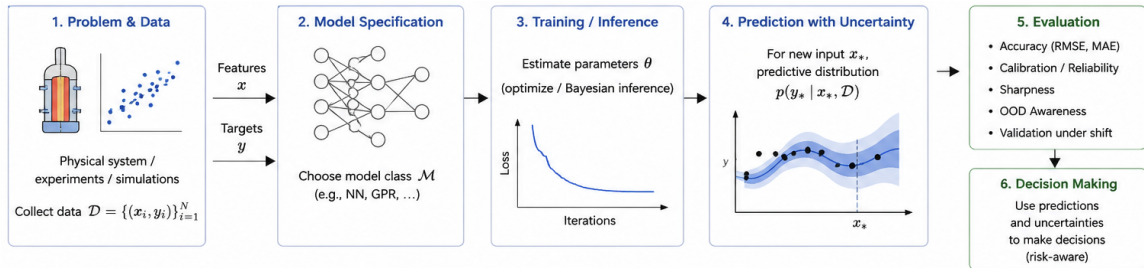
Part III

UQ for Neural Networks

4. Uncertainties in an ANN

5. Posterior Predictive Distribution

6. Integrate UQ with ANNs

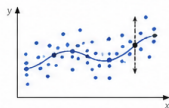


Outline:

- Review sources of uncertainty
- Predictive distribution of QoIs
- Bayesian neural networks and alternative methods
- Trustworthy UQ
- Decision making under uncertainty

SOURCES OF UNCERTAINTY

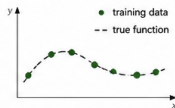
1 Noisy observations (irreducible)



Outcome variability even if the true function is known.

Mitigated by:
Better measurements,
noise modeling

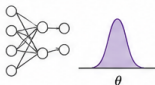
2 Finite training data (data scarcity)



Limited data leads to uncertainty about the true function.

Mitigated by:
More data, data augmentation,
active learning

3 Uncertain parameters (within model)



Many parameter values θ are plausible given the data. (e.g., posterior over θ)

Mitigated by:
Bayesian inference,
regularization

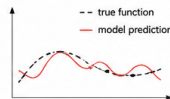
4 Model architecture (model class)



Different model families can explain the data similarly but predict differently.

Mitigated by:
Model comparison,
ensembles over architectures

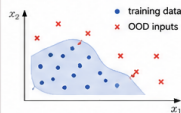
5 Approximation error (model limitations)



Model class cannot represent the true function well (bias / misspecification).

Mitigated by:
Richer models, physics-informed models, hybrid approaches

6 Out-of-distribution inputs (OOD)



Predictions are uncertain when inputs are outside the training distribution.

Mitigated by:
OOD detection, robust models,
collect data in relevant regions

Aleatoric (data) uncertainty

Epistemic (model) uncertainty

- Total predictive uncertainty can be from a combination of multiple sources.
- Different methods capture different components of it.
- UQ for predictive models (including NNs) can be more involved than the standard UQ.

4. Uncertainties in an ANN

5. Posterior Predictive Distribution

6. Integrate UQ with ANNs

Gaussian Process Regression (GPR) [12]:

$$y(\mathbf{x}) = f(\mathbf{x}) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \Sigma_\varepsilon)$$

where

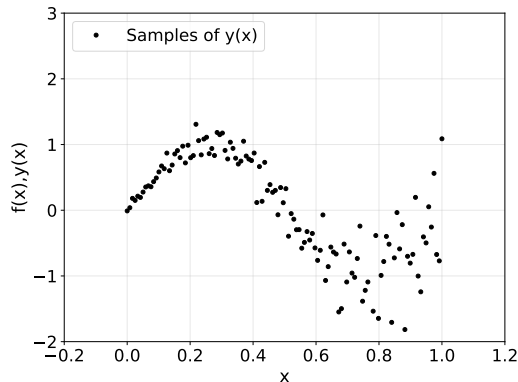
$$f(\mathbf{x}) \sim \mathcal{N}(\mu(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'; \boldsymbol{\theta})),$$

with the **mean** and **covariance** of $f(\mathbf{x})$:

$$\begin{aligned}\mu(\mathbf{x}) &= \mathbb{E}[f(\mathbf{x})] \\ k(\mathbf{x}, \mathbf{x}'; \boldsymbol{\theta}) &= \text{cov}[f(\mathbf{x}), f(\mathbf{x}')] \end{aligned}$$

1. Given noisy data $\mathcal{D}$, estimate or infer $\boldsymbol{\theta}$.
2. Make predictions at test samples $\mathbf{x}^*$.
3. **Posterior Predictive Distribution:**

$$p(\mathbf{y}^* | \mathbf{x}^*, \mathcal{D}) = \int_{\boldsymbol{\theta}} p(\mathbf{y}^* | \mathbf{x}^*, \boldsymbol{\theta}, \mathcal{D}) p(\boldsymbol{\theta} | \mathcal{D}) d\boldsymbol{\theta}$$



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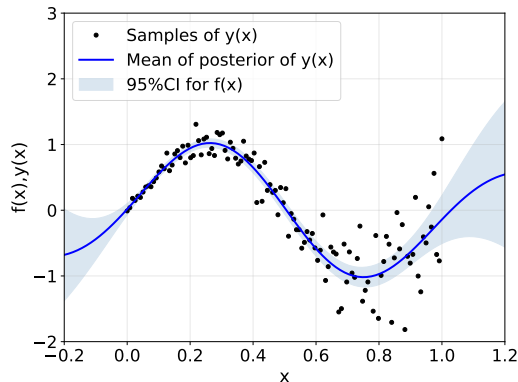
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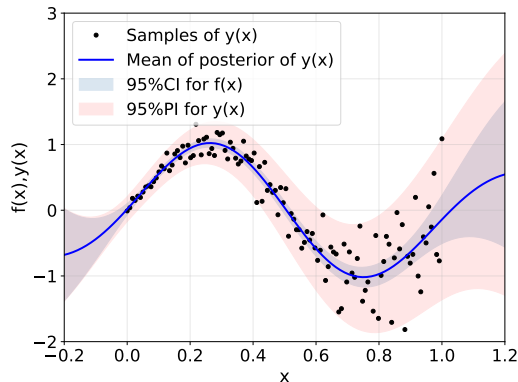
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4. Uncertainties in an ANN

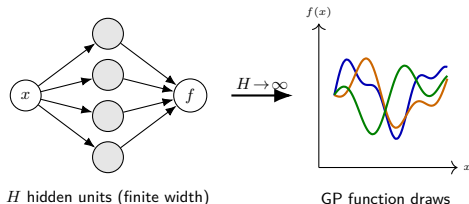
5. Posterior Predictive Distribution

6. Integrate UQ with ANNs

Put a **prior** on the parameters $\theta = \{w, b\}$ of a **single-hidden-layer** network:

$$f(x) = \sum_{j=1}^H w_j^{(2)} \phi(w_j^{(1)} x + b_j),$$

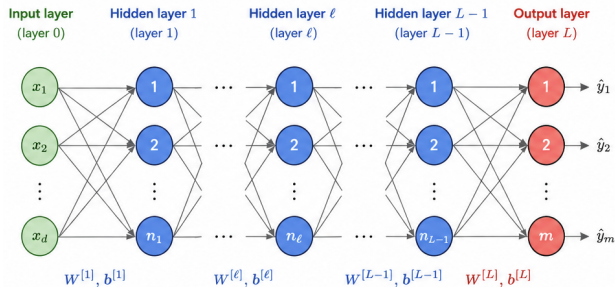
$$w_j^{(1,2)} \sim \mathcal{N}(0, \sigma_w^2), b_j \sim \mathcal{N}(0, \sigma_b^2)$$



- H = number of hidden units (the layer's *width*; *not* the number of layers)
- Neal [10]; Ch.2.: as $H \rightarrow \infty$ (wider, not deeper), $f(x)$ becomes a Gaussian process
- Each hidden unit is one random “basis function”; infinitely many i.i.d. terms
 $\Rightarrow$ Central Limit Theorem $\Rightarrow f(x)$ is Gaussian at every x
- σ_b^2 enters the induced kernel too, controlling the size of an x -independent offset.

Idea: A Bayesian Neural Network (BNN) with *finite* H is a flexible and scalable cousin of a GP.
 $\Rightarrow$ same machinery, different function class!

- Data: $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^n$
- Features: $\mathbf{x}_i \in \mathbb{R}^d$; Output: $\mathbf{y}_i \in \mathbb{R}^m$
- ANN: $f_{\theta} : \mathbb{R}^d \rightarrow \mathbb{R}^m$
- Parameters: $\theta = \{\mathbf{W}_l, \mathbf{b}_l\}_{l=1}^L$

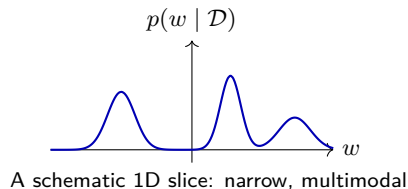


Bayesian Neural Network (BNN):

1. Construct the likelihood $p(\mathcal{D}|\theta)$
 2. Apply the Bayes rule to infer $p(\theta|\mathcal{D})$, ideally using, an MCMC method
 3. Infer posterior & posterior predictive distributions $p(\mathbf{y}^*|\mathbf{x}^*, \mathcal{D})$
- ✔ **With an explicit observation/likelihood model, predictive uncertainty may combine parameter (θ) uncertainty and aleatoric uncertainty.**

- θ can have 10^4 – 10^9 dimensions.
- Likelihood is a deep composition of non-linearities $\Rightarrow$ non-conjugate prior/posterior.
- Posterior is typically **multimodal**: permuting hidden units gives equally good, symmetric solutions.
- $p(\mathcal{D}) = \int p(\mathcal{D} \mid \theta) p_0(\theta) d\theta$ can be **intractable**!
- No Cholesky trick like in GP regression.

$\Rightarrow$ We must **approximate** $p(\theta \mid \mathcal{D})$.



Some approximate methods with lower computational costs:

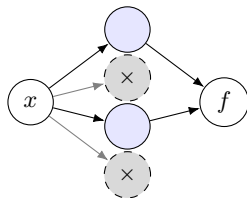
1. Monte Carlo (MC) Dropout
2. Laplace Approximation
3. Deep Ensembles
4. Variational Inference (Bayes by back-propagation)

✔ None of these is automatically **trustworthy**: validate coverage, calibration and behaviour under distribution shift/out-of-distribution (OOD) inputs.

- **Dropout**: randomly zero hidden units with some probability p° during training (a standard regularizer) $\rightarrow$ Keep dropout active only at test.
- Gal & Ghahramani, 2016 [4]: leaving dropout **switched on at test time** and averaging T stochastic forward passes approximates the BNN predictive distribution:

$$\hat{\mu}(x^*) = \frac{1}{T} \sum_{t=1}^T f_{\theta_t}(x^*), \quad \hat{\sigma}^2(x^*) = \mathbb{V}_t[f_{\theta_t}(x^*)]$$

- Each mask $\approx$ one sample from an implicit $q(\theta)$. But dropout has no notion of “dropping” a bias, so in practice **biases stay deterministic; only the effective weights are randomized**.
- **Very cheap**: reuses code you already have!
- Caveat: often **underestimates uncertainty**; quality depends heavily on p° and network width .



One dropout mask = One sampled network

Laplace Approximation

- Read the recent review [1] and reference therein.
- Fit a Gaussian distribution using the MAP estimates for $\theta = \{\mathbf{w}, \mathbf{b}\}$:

$$\begin{aligned} p(\theta|\mathcal{D}) &\approx \mathcal{N}(\theta_{\text{MAP}}, \mathbf{H}^{-1}), \\ \mathbf{H} &= -\nabla_{\theta}^2 \log p(\theta|\mathcal{D})|_{\theta_{\text{MAP}}} \end{aligned}$$

- The Hessian $\mathbf{H}$ has only row/column per scalar parameter.
- Computationally cheap
- Caveat: Usually a poor global fit in high dimensions.

Deep Ensembles

- Proposed by Lakshminarayanan *et al.* 2017 [9].
- Train M models independently (random initialisation/data order, etc)
- Make prediction with each model as usual
- Empirically find the mean and variance of the QoIs from M models
- Caveat: Between-model variance is not a calibrated posterior variance and may under- or over-estimate predictive uncertainty.
- Caveat: Often strongly biased (low convergence with M).

- Proposed by Blundell *et al.* 2015 [3] with basics in 90's [7].
 - Can be applied to both weights and biases, $\theta = \{\mathbf{w}, \mathbf{b}\}$.
 - The idea of variation inference is turning inference into an optimisation problem.
1. Pick a tractable family $q_{\beta}(\theta)$ (e.g. diagonal Gaussian, $\beta = (\mu, \sigma)$) and minimize $\text{KL}(q_{\beta} \parallel p(\theta|\mathcal{D}))$. This is equivalent to maximizing the Evidence Lower Bound (ELBO):

$$\mathcal{L}(\theta) = \underbrace{\mathbb{E}_{q_{\beta}(\theta)} [\log p(\mathcal{D} \mid \theta)]}_{\text{fit the data}} - \underbrace{\text{KL}(q_{\beta}(\theta) \parallel p_0(\theta))}_{\text{stay close to prior}}$$

2. For a mean-field Gaussian posterior on a single scalar parameter θ_i (a weight or a bias), $q(\theta_i) = \mathcal{N}(\mu_i, \sigma_i^2)$ and prior $p_0(\theta_i) = \mathcal{N}(0, \sigma_p^2)$, the Kullback–Leibler (KL) term is closed-form:

$$\text{KL}(q \parallel p) = \log \frac{\sigma_p}{\sigma_i} + \frac{\sigma_i^2 + \mu_i^2}{2\sigma_p^2} - \frac{1}{2}$$

3. Summed over every θ_i (weight or bias) to get the full KL term. Only the data term needs Monte Carlo estimation.

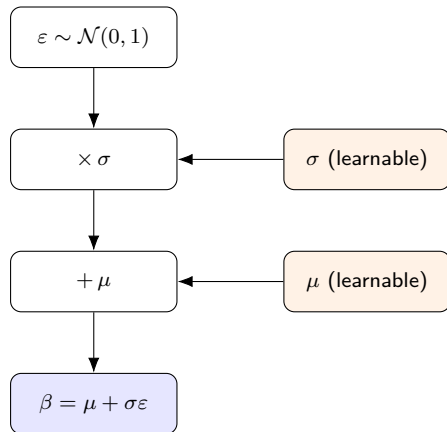
We can do Bayes by Back-propagation using the reparameterization trick.

Applies to *every* scalar parameter (weight or bias).

We cannot back-propagate through a *sample* $\beta \sim q_{\beta}(\theta)$ directly. Instead, write the sample as a deterministic function of noise:

$$\varepsilon \sim \mathcal{N}(0, 1), \quad \beta = \mu + \sigma \varepsilon$$

- μ, σ become ordinary learnable parameters
- In practice $\sigma = \log(1 + e^{\rho})$ (softplus) so $\sigma > 0$ for any $\rho \in \mathbb{R}$
- Gradients of $\mathcal{L}$ w.r.t. μ, ρ flow through the sampled network via ordinary backprop



Part IV

Trustworthy UQ, Risk Assessment, and Decision Making

7. Trustworthy UQ for Predictive Models

8. UQ for Risk Assessment and Decision Making

Confidence Interval (CI)

Uncertainty in an **estimated quantity**, e.g.

$$\mu(\mathbf{y}) = \mathbb{E}[\mathbf{y} \mid \mathbf{x}].$$

A frequentist statement about repeated procedures.

Prediction Interval (PI)

Uncertainty in a **future observation**, e.g.

$$\mathbf{y}_{\text{new}} \mid \mathbf{x}, \mathcal{D}.$$

Usually wider: includes response uncertainty and observation variability.

Credible Interval

A Bayesian posterior probability statement, e.g.

$$p(y \in [a, b] \mid \mathcal{D}) = 0.95.$$

Depends on prior, likelihood and data.

✔ A “95% interval” is **incomplete** unless we say:

- 95% interval for what, and
- under which probabilistic interpretation.

How to Make Predictive Uncertainty Trustworthy?

Concept	Question	Possible Measure
Accuracy	Is the central prediction close to the truth?	MAE or $\text{RMSE} = \sqrt{\sum_{i=1}^N (y_i - \hat{y}_i)^2 / N}$
Empirical Coverage	How often does the predictive interval [†] contain the observation?	$\hat{C}_\alpha = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[y_i \in C_\alpha(x_i)]$ at a specific α
Calibration/ Reliability	Do nominal predictive probabilities match observed frequencies?	$\hat{C}_\alpha = \alpha$ with multiple α
Sharpness	How narrow/wide are the prediction intervals?	Mean interval width $\text{MIW}_\alpha = \frac{1}{N} \sum_{i=1}^N [u_\alpha(x_i) - \ell_\alpha(x_i)]$
OOD Awareness	Does uncertainty respond appropriately away from the training distribution?	OOD detection/ uncertainty shift

✔ **A trustworthy UQ model should be accurate, calibrated, sharp, and aware of distribution shift.**

[†] If only a CI is available or CI and PI coincide under the adopted predictive model, the same diagnostics may be applied with that interpretation stated explicitly.

In ML-based UQ, the model may output a predictive distribution:

$$p(y^* | x^*, \mathcal{D}),$$

or a nominal 90% prediction interval:

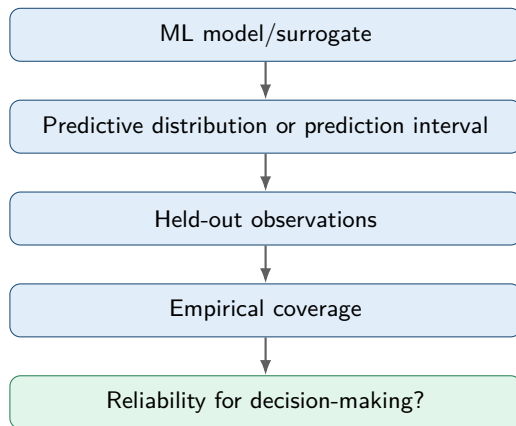
$$C_{0.9}(x^*) = [\ell_{0.9}(x^*), u_{0.9}(x^*)].$$

The calibration question is then empirical:

$$p(y \in C_{0.9}(x)) \stackrel{?}{=} 0.9.$$

On a **validation (held-out) set**:

$$\hat{C}_{0.9} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[y_i \in C_{0.9}(x_i)].$$



✓ **The uncertainty estimate is itself an object that must be validated.**

Actual Meaning of a 90% Predictive Interval

For an NTH surrogate predicting peak cladding temperature,

$$\hat{T}_{\text{PCT}}(\mathbf{x}),$$

suppose the model reports an empirical 90% predictive interval

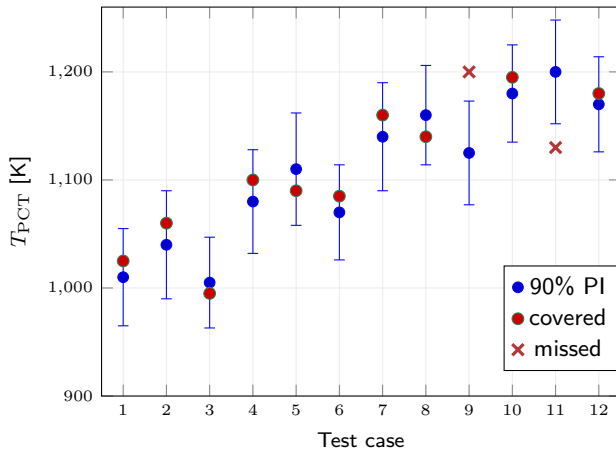
$$\hat{C}_{0.9}(\mathbf{x}) = [\ell_{0.9}(\mathbf{x}), u_{0.9}(\mathbf{x})].$$

Population-level meaning: For multiple new cases (X, Y) drawn from the target population,

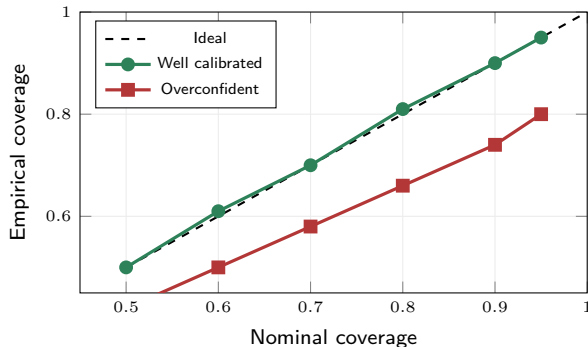
$$p(Y \in C_{0.9}(X)) \stackrel{?}{=} 0.9.$$

Meaning: Across repeated/new realizations from that population, 90% of realised outcomes fall within their corresponding predictive intervals.

✔ A narrow interval is not necessarily a good interval.



Empirical 90% PI $\nRightarrow$ Nominal 90% PI automatically.



Calibration asks whether nominal probabilities have empirical meaning:

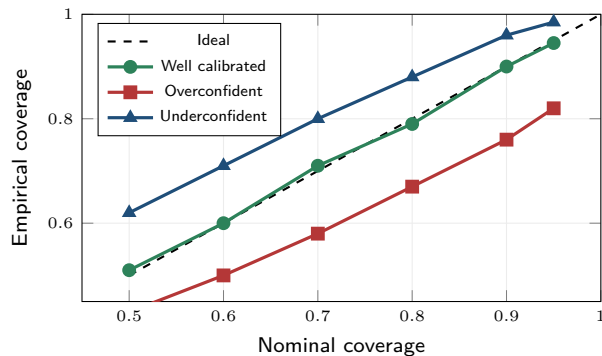
$$\hat{C}_\alpha \approx \alpha.$$

Trying multiple values of α .

Note that:

- A model can be accurate but overconfident.
- A model can be calibrated but too conservative.
- A model can be calibrated on average but fail near a safety boundary.

✓ This is why predictive intervals lead naturally to reliability diagrams, coverage tests and regime-wise validation.



	Coverage	Mean PI width
Model A	90%	300 K
Model B	90%	80 K

Calibration: For several nominal levels α :

$$\hat{C}_\alpha = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[y_i \in C_\alpha(\mathbf{x}_i)].$$

Sharpness: How wide/narrow is the PI?

Perfect coverage can be achieved with uselessly wide intervals (low sharpness).

$$\text{MIW}_\alpha = \frac{1}{N} \sum_{i=1}^N [u_\alpha(x_i) - \ell_\alpha(x_i)]$$

Assess both:

$$\boxed{\text{Calibration/reliability}} + \boxed{\text{Sharpness}}$$

✓ **Good uncertainty is calibrated *and* informative.**

Suppose the test-set result is

$$\hat{C}_{0.9} = 0.90.$$

Looks excellent until we **condition on operating regime**:

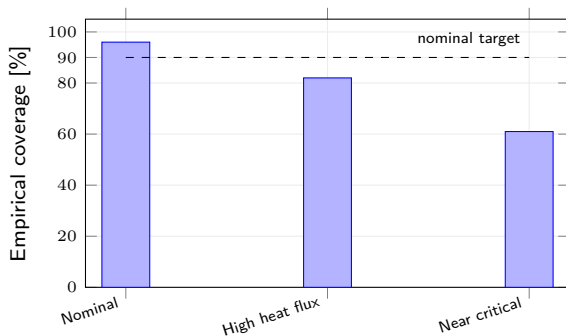
Operating regime	90% PI coverage
Nominal conditions	96%
High heat flux	82%
Near critical regime	61%

Marginal calibration:

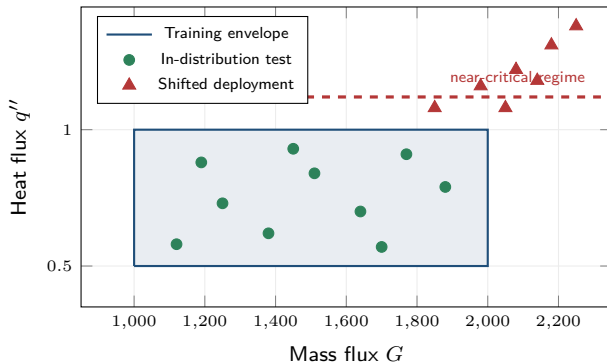
$$p(Y \in C_\alpha(X)) = \alpha$$

does **not** imply local/conditional calibration:

$$p(Y \in C_\alpha(X) \mid X = x) = \alpha.$$



✔ **Average calibration can conceal failure where it matters most.**



Past calibration does not guarantee future reliability.

Validation on the original distribution,

$$(\mathbf{x}, y) \sim p_{\text{train}},$$

may give

$$\hat{C}_{0.9} = 0.91.$$

But deployment may occur under

$$p_{\text{deploy}}(\mathbf{x}) \neq p_{\text{train}}(\mathbf{x}),$$

for example at higher heat flux or in a new flow regime.

Then the same nominal 90% interval may yield

$$\hat{C}_{0.9} = 0.64.$$

✓ Calibration is distribution-dependent: validate uncertainty where the model will actually be used.

7. Trustworthy UQ for Predictive Models

8. UQ for Risk Assessment and Decision Making

Suppose an ML surrogate predicts

$$\hat{T}_{\text{PCT}} = 1170 \text{ K},$$

with safety threshold

$$T_{\text{crit}} = 1200 \text{ K}.$$

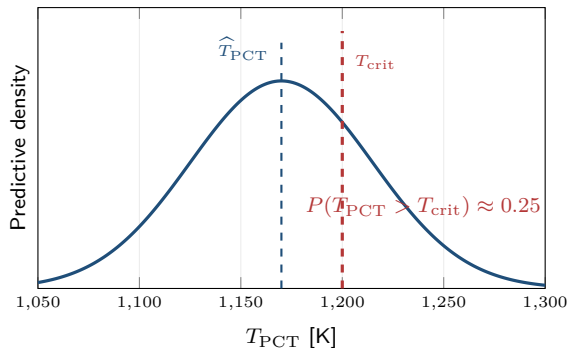
A point-prediction view says

$$1170 < 1200 \Rightarrow \text{"safe"}.$$

But UQ gives $p(T_{\text{PCT}} \mid \mathbf{x}, \mathcal{D})$,

$\Rightarrow$ A **decision-relevant quantity** is:

$$p(T_{\text{PCT}} > T_{\text{crit}} \mid \mathbf{x}, \mathcal{D})$$



Is this safe?

- ✔ UQ quantifies a risk-relevant probability; but it does NOT define acceptable risk.

✔ Instead of “What is the prediction?”, ask **“What is the probability of a decision-dependent adverse outcome?”**

- Assume we have obtained predictive distribution $p(Y|\mathcal{D})$ for an **uncertain outcome** Y .
- Let $a \in \mathcal{A}$ denote an **action** and $L(a, Y)$ the **loss** if uncertain outcome Y occurs.
- A **Bayesian decision rule** can choose an optimal action:

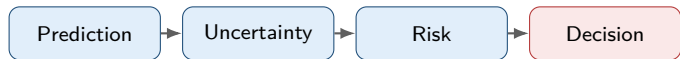
$$a^* = \arg \min_{a \in \mathcal{A}} \mathbb{E}[L(a, Y) \mid \mathcal{D}] = \arg \min_{a \in \mathcal{A}} \sum_Y L(a, Y) p(Y|\mathcal{D})$$

In safety-critical systems, losses are asymmetric:

$$L(\text{continue operation, unsafe}) \gg L(\text{protective action, safe}).$$

Action	System remains safe	Safety limits exceeded
Continue operation	Low/no loss	Catastrophic loss
Protective actions	Operational cost	Reduced consequences

- ✔ The same predictive probability can lead to different actions under different consequences.

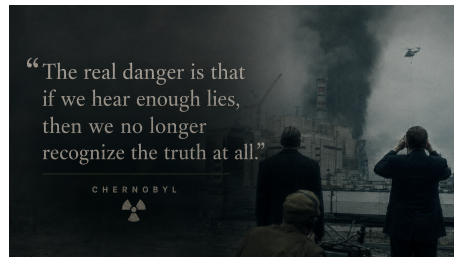


Even a perfectly calibrated estimate such as

$$p(\text{failure}) = 10^{-3}$$

does NOT determine whether that risk is acceptable.

- ✓ Predictive uncertainty is only one input to a decision; acceptable risk depends on consequences, constraints and policy.



- UQ is not about attaching error bars to predictions.
- UQ is about representing:
 - What is uncertain.
 - Why it is uncertain.
 - Whether that uncertainty is adequate for decision-making.
- UQ analyses can be involved. It requires expertise and understanding the problem and purpose.
- UQ for predictive models is even more involved!
- Note that uncertainty itself should be validated (trustworthy UQ).

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